PROFILE HANDBOOK

Programs A-I

The cartridge-free field manual for Power Glove Vision profiles.

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How Bad Street Brawler configured the original Power Glove - and how PowerGlove Vision recreates the useful behaviour with a camera.

The wonderfully strange original idea

Bad Street Brawler contained nine configuration programs labelled A through I. The player loaded one into the Power Glove, switched off the NES, swapped to a different cartridge within roughly 30 seconds, and kept using the downloaded mapping.

These programs were not replacement firmware. They were small configurations for the glove's resident gesture interpreter. Each combined hand position, depth, wrist angle, and finger flex into ordinary NES controller inputs. The next game therefore did not need special Power Glove support.

> **POWERGLOVE VISION CHANGES THE WORKFLOW** The UNO Q keeps all nine profiles ready at once. Select one on the Dashboard or let RetroPie choose it when a game launches. Bad Street Brawler never has to be opened first.

Quick selector

Program	Best fit	Camera-era personality
A	Pinball	Two finger flippers, wrist tilt, combined-flipper mode
B	Joust	Steer by position; curl a finger to flap
C	Gyruss	Rotate by wrist angle; fire and bomb gestures
D	Challenge mode	Reversed directions with thumb/index buttons
E	Defender II	Ship movement, fire, smart bomb, evasive movement
F	Sesame Street 1-2-3	Open-hand Yes and closed-hand No
G	Gun Smoke	Walk by position; aim by wrist angle
H	Training / general play	Familiar controls with pulsed buttons
I	Knight Rider / driving	Wrist steering, throttle, brake, and turbo

Menu gestures shared by every program

Gesture	See it	Controller result
Hold a V sign for about 0.7 seconds		Start or pause
Hold a thumbs-up with the other fingers closed for about 0.7 seconds		Select

These menu poses briefly suppress directions and action gestures while they form. Return to a relaxed open hand after the command is recognized.

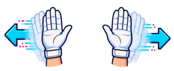
Program cards

A - Pinball rig

Gesture	See it	Controller result
Curl the index finger		Right flipper / A
Curl the thumb		Left flipper / Up
Rotate the wrist toward six o'clock		Tilt / B
Pull the hand away from the camera		Toggle combined-flipper behaviour

Use it for: pinball tables and games that benefit from two independent finger actions.

B - Joust rig

Gesture	See it	Controller result
Move the hand left or right		Steer left or right
Curl the thumb		Turn the rider
Curl the index or middle finger		Pulsed flap input

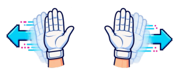
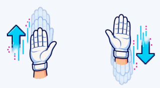
Use it for: Joust and any game where rhythmic, repeated presses matter.

C - Gyruss rig

Gesture	See it	Controller result
Roll the wrist left or right		Rotate counter-clockwise or clockwise
Keep the index finger straight		Continuous fire
Pull the hand away from the camera		Launch a bomb

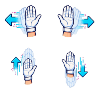
Use it for: circular shooters and games with rotation plus rapid fire.

D - Mirror-world rig

Gesture	See it	Controller result
Move the hand left or right		Reversed right or left direction
Raise or lower the hand		Reversed down or up direction
Curl the thumb		First action button
Curl the index finger		Second action button

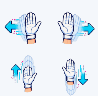
Use it for: deliberate chaos, party challenges, or accessibility experiments that need inverted direction mappings.

E - Defender rig

Gesture	See it	Controller result
Move the whole hand		Move the ship
Curl the thumb		Fire
Rotate the wrist toward six o'clock		Smart bomb
Curl the ring finger		Rapid horizontal movement

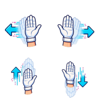
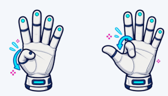
Use it for: Defender II and multi-action shooters.

F - Yes / No rig

Gesture	See it	Controller result
Close every finger into a fist		No
Move an open hand in any direction		Yes

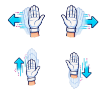
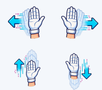
Use it for: Sesame Street 1-2-3 and simple choice-driven games.

G - Gun Smoke rig

Gesture	See it	Controller result
Move the whole hand		Walk using X, Y, and depth movement
Curl the index finger		Fire
Roll the wrist		Choose firing direction
Curl the thumb and ring finger		Suppress action for menu navigation

Use it for: Gun Smoke and shooters with movement plus directional fire.

H - Training rig

Gesture	See it	Controller result
Move the whole hand		Conventional directions
Curl the thumb		First pulsed action button
Curl the index finger		Second pulsed action button
Return the hand to center		Training acknowledgement

Use it for: learning the system or giving an unmapped game a sensible general-purpose starting point.

I - Driving rig

Gesture	See it	Controller result
Roll the wrist left or right		Steer
Curl the index finger		Throttle
Lower the hand		Brake
Push the hand toward the camera		Turbo
Curl the thumb		Auxiliary action

Use it for: Knight Rider and driving games that need steering, speed, and one extra action.

How PowerGlove Vision selects a program

Manual selection is available on the UNO Q Setup page. Automatic selection uses the exact, case-insensitive ROM basename in `/etc/powerglove/games.json`. RetroPie's launch hook authenticates a profile request, the UNO Q releases any held controls, changes mapping, recalibrates, and acknowledges the new profile on its blue matrix.

```
JSON
{
  "Joust (USA).zip": "program_b",
  "Gyruss (USA).nes": "program_c",
  "Gun.Smoke (USA).7z": "program_g"
}
```

The matrix displays `A` through `I`. Unknown games turn gesture control off, preventing a previous title's mapping from leaking into the next one.

Practise safely

1. Open `http://UNO-Q-NAME.local:8088/learn`.
2. Keep the whole hand visible and select **Calibrate**.
3. Move slowly until the intended gesture is recognized consistently.
4. Open Debug to compare hand motion with generated controller output.
5. Start controller delivery only when ready to play.

Learn mode works without RetroPie and automatically starts the camera while suppressing controller output. This also works while **Gestures off** is the selected profile. Leaving Learn restores the selected mode, so it is a safe place to build muscle memory without changing the saved or active selection.

What remains special

The nine profiles emit useful conventional controller inputs, so ordinary NES cores can use them today. Native Power Glove packets would still matter for glove-aware software such as Super Glove Ball and Bad Street Brawler's glove-only zap. PowerGlove Vision preserves the richer analogue and finger data needed for that future emulator integration; it does not modify ROM images.

For installation, secure pairing, RetroArch mapping, and troubleshooting, see `INSTALL_README.md`.

Project note

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